

FOURTH SEMESTER

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	Python Programming				
Code	BCAN14201				
Course Type	DSC	L	T	P	Credit
Pre-Requisite		3	1	0	4
Course Objectives	This subject provides in-depth knowledge of developing and debugging Python Programs using core data structures like Lists, Dictionaries, Tuples, and Strings as well as understand the concept of functions, modules and NumPy Library.				
Course Outcomes					
CO1	To Acquire programming skills in core Python using various programming constructs				
CO2	To Implement Python programs using functions and strings.				
CO3	To Implement methods to create and manipulate lists, tuples, and dictionaries.				
CO4	For Implementation of Python Library: NumPy.				
Module	Course Contents			Contact Hrs.	Mapped CO
1	Overview of programming: structure of a python program, elements of python; Introduction to python: python interpreter, using python as calculator, python shell, indentation. Atoms, identifiers and keywords, Comments, literals, strings, operators, Creating python programs: input and output statements; control statements (branching, looping, conditional statement, exit function, difference between break, continue and pass); errors and exceptions; Iteration and recursion: conditional execution, alternative execution, nested conditionals, the return statement.			15	CO1
2	Conditional and Looping Construct, Functions: if - else statement and nested if – else while, for, use of range function in for, Nested loops, break, continue, pass statement, Use of compound expression in conditional constructs; Module(Importing entire module or selected objects using from statement), Functions from math, random, time & date module, Defining and invoking functions: Built-In Function, user defined function, default arguments, passing parameters (default parameter values, keyword arguments); Scope of variables, void functions and functions returning values.			15	CO2

3	Strings: Introduction to String, working with Text, Creating Strings of Characters, Using Special Characters in Strings, creating a Multiline String, String functions concepts & their use; String operators: +, *, in, not in, range, indexing, slicing; String built-in functions & methods: len, capitalize, find, isalnum, isalpha, isdigit, lower, islower, isupper, upper, lstrip, rstrip, isspace, istitle, partition, replace, join, split, count, decode, encode, swap case, Strings constants; Lists, Tuples: Concept of mutable lists, creating, initializing and accessing the elements of list, List operations (Concatenation, Repetition, Membership, slicing); List comprehensions, List functions & methods: len, insert, append, extend, sort, remove, reverse, pop, Immutable concept, creating, initializing and accessing the elements in a tuple; Tuple functions.	15	CO3
4	Sets and Dictionaries: Concept of Sets, creating, initializing and accessing the elements of sets, Concept of key-value pair, creating, initializing and accessing the elements in a dictionary, Traversing, appending, updating and deleting elements; Introduction of NumPy: Installing NumPy, Array creation and printing Array ndim, shape, size, dtype, itemsize and data.	15	CO4

Suggested Readings

1. Kenneth A. Lambert, the Fundamentals of Python: First Programs, Cengage Learning, ISBN: 978-1111822705.
2. Practical Programming: An introduction to Computer Science Using Python, second edition, Paul Gries, Jennifer Campbell, Jason Montojo, The Pragmatic Bookshelf.
3. NumPy for Beginners: first Step to Learn to Data Science – Preeti Saraswat.
4. David Beazley, Brian K. Jones “Python Cookbook”. O’Reilly Publications.

Online Resources

1. <https://nptel.ac.in/courses/106106145>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2		2		2	1	2		1			2	1	1
CO2	2		2		2	2	2		1			2	1	1
CO3	2	2	2	2	3	2	3		2	2	2	2	2	2
CO4	2	2	3	2	2	2	3		3	2	2	2	3	3

Program	Bachelor of Computer Applications					
Year	II	Semester		IV		
Course Name	.Net Framework & C#					
Code	BCAN14202					
Course Type	DSC	L	T	P	Credit	
Pre-Requisite		3	1	0	4	
Course Objectives	The Subject provides the Fundamental Concepts of Windows Desktop Application and Website Development with machine learning and data science through .Net framework and C#.					
Course Outcomes						
CO1	To develop an understanding of .Net technology using C# and Asp.net.					
CO2	To understand the Database Connectivity.					
CO3	To develop an understanding of Static and Dynamic web pages.					
CO4	To understanding API, REST, SOAP and AJAX					
Module	Course Contents			Contact Hrs.	Mapped CO	
1	.Net Framework: Introduction and Origin of .Net technology; Framework Components, Common Language Runtime (CLR) and FCL; Managed and Unmanaged Code; Common Type System (CTS) & Common Language Specification (CLS); Microsoft Intermediate Language (MSIL) and Metadata; Just-In-Time Compilation (JIT); Garbage Collection; Base Classes and Ms.Net Namespaces. Object and Classes: Properties (Read, Write), Indexers, Inheritance (Multilevel and Hierarchical), Constructor Polymorphism (Runtime, Compile Time), Operator Overloading, Interfaces, Delegates and Events, Boxing and Un-boxing.			15	CO1	
2	C# Basics: Introduction and Evolution of C# , Types, Identifiers, Tokens, Literals, Type Conversion and Casting, Operators, Data Structures in C#: Enum, Arrays, ArrayList, Strings; Control Statements and Looping: If Statement, Switch Statement, For Loop, While Loop, Do While Loop, For each Loop; C# Libraries and Assemblies: Input-output (Streams Classes); Multithreading; Networking and Sockets; Managing Console I/O Operations; .NET Assemblies: Type of Assemblies, GAC(Global Assembly Cache), Concept of Strong Names.			15	CO2	
3	Windows and Website Development: Windows Forms (A Skeletal Form Based Windows Program, Remoting: Server Activated Object, Client Activated Object; Marshalling: Marshal by value, Marshal by reference; Debugging, Exceptions and Error Handling; ASP.NET Web Form Controls: User controls and Server Controls; Web Services: UDDI, DISCO, WSDL; ADO.NET: Architecture, Difference between Dataset and data Reader, Connection and Command Object; Distributed applications; Reflection; Globalization and Localization; Authentication and Authorizations; XML in .NET.			15	CO3	
4	Advanced Concepts: REST AND SOAP: Rest, Restful, Soap, WCF, WPF, Implementation of Rest and Soap, Restful Vs Soap. Web server: web server, types, web server used in .net; Ajax Controls: AJAX and need for AJAX, Implement with JavaScript, ASP.NET AJAX – Update Panel, Update Progress etc., ASP.NET Ajax Control toolkit, Client-side Template Rendering–Data View control.			15	CO4	

Suggested Readings

1. Balagurusamy," Programming. with C#", Tata McGraw Hill Publication.
2. Stephen C. Perry, Atul Kahae, Stephen Walther, Joseph Mayo," Essential of .NET and Related Technologies with a focus on C#, XML, ASP.net and ADO.net", Pearson.
3. Joseph Albahari, "C# 8.0 Pocket Reference", O'Reilly.

Online Resources

1. <https://archive.nptel.ac.in/courses/605/607/608/609>
2. <https://archive.nptel.ac.in/courses/703/704/705/706>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	2	1	2	1					2	1	1
CO2	2	2	2	2	2	2	1		2	2		2	2	2
CO3	2	2	1	2	3	2	2		2	2	2	2	2	2
CO4	2	2	1	2	3	2	2		2	2	2	3	3	3

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	Basics of Design & Analysis of Algorithms				
Code	BCAN14203				
Course Type	DSC	L	T	P	Credit
Pre-Requisite		3	0	0	3
Course Objectives	To know the importance of studying the complexity of a given algorithm and various design techniques. Utilizing data structures and/or algorithmic design techniques in solving new problems. Understanding basic computability concepts and the complexity.				
Course Outcomes					
CO1	Able to Argue the correctness of algorithms using inductive proofs and analyze worst-case running times of algorithms using asymptotic analysis.				
CO2	Able to explain important algorithmic design paradigms (divide-and-conquer, greedy method) and apply when an algorithmic design situation calls for it.				
CO3	Able to explain important algorithmic design paradigms (dynamic-programming and Backtracking) and apply when an algorithmic design situation calls for it.				
CO4	Able to Explain the major graph algorithms and Employ graphs to model engineering problems, when appropriate.				
Module	Course Contents			Contact Hrs.	Mapped CO
1	Basic Concepts of Algorithms: Definition of algorithm; Characteristic of algorithm; Pseudo Codes & Time Complexity of Basic Control Structures; Time and Space Complexity of Insertion Sort; Selection Sort; Heap Sort; Bubble Sort; Asymptotic Notations (Growth of Functions).			12	CO1
2	Divide and conquer: Binary Search, Maximum & Minimum, Merge Sort, Quick Sort, Strassen's matrix multiplication; Greedy Method: General method, Knapsack Problem, Travelling Salesman problem, Job Sequencing with deadline, Optimal Storage on tapes, Huffman Codes, An Activity Selection Problem.			12	CO2
3	Dynamic Programming: Assembly Line Scheduling, Matrix Chain Multiplications, Longest Common Subsequence; Backtracking: General method, N Queens Problem, Sum of subsets, Hamiltonian Circuit Problem.			12	CO3
4	Branch & Bound: Introduction, Live Node, Dead Node and Bounding Functions, Knapsack Problem, Assignment Problem; Analysis of Graph Algorithms: Elementary Graph Algorithms, Multistage Graphs, Minimum Spanning Trees: Kruskal's & Prim's Algorithm, Single Source Shortest Path: Dijkstra's & Bellman Ford.			12	CO4

Suggested Readings

1. Thomas H. Cormen, "Introduction to Algorithms", MIT Press.
2. Horowitz & Sahani, "Fundamentals of Algorithms", Galgotia Publications.
3. Aho, Ullman, "Design & Analysis of Computer Algorithms", Pearson.
4. Johnsonbaugh, "Algorithms", Pearson.
5. Bressard, "Fundamentals of Algorithms", PHI.

Online Resources

1. <https://archive.nptel.ac.in/courses/106/106/106106131/>.
2. https://onlinecourses.nptel.ac.in/noc19_cs47/preview

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C01	2	1	2	2		3	1		1	2	1	3	3	3
C02	2	2	2	3		3	1		1	2	1	3	3	3
C03	2	2	2	3		3	1		1	2	1	3	3	3
C04	2	2	2	3		3	1		1	2	1	3	3	3

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	Data Mining				
Code	BCAN14211				
Course Type	GE	L	T	P	Credit
Pre-Requisite		3	1	0	4
Course Objectives	This course provides an in-depth exploration of data mining and data warehousing techniques, methodologies, and applications. Students will learn how to extract valuable insights from large datasets, design and implement data warehouses, and apply data mining algorithms for knowledge discovery.				
Course Outcomes					
CO1	To understand the basic concept Data Warehousing and Data Mining				
CO2	To understand the concept of preprocessing, OLAP and Frequent pattern Mining				
CO3	To understand the concept of Classification				
CO4	To understand the concept of Clustering				
Module	Course Contents			Contact Hrs.	Mapped CO
1	Introduction to Data Mining and Data Warehousing: Overview of data mining and knowledge discovery process Role and importance of data warehouses, Key concepts and components of data mining and data warehousing; Multi-Dimensional Data Model: Introduction, Elements, steps in dimensional modeling, Multi-Dimensional Schema; Data Warehouse Architecture: The 3-Tier Data Warehouse Architecture, The Bus Architecture.			15	CO1
2	Data Preprocessing: Overview, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Discretization; Data Warehouse Modeling: Data Cube, Typical OLAP Operations, Role of Concept Hierarchies, OLAP Server Architectures; Mining Frequent Patterns: Basic concepts Frequent Item set mining method: the Apriori Algorithm, Generating Association Rules from frequent item sets, FP Growth Algorithm.			15	CO2
3	Classification: General Approach to solving classification problems; Classification by decision Tree Induction: Attribute selection measure, Tree pruning; Bayesian Classification: Bayes’ Theorem; Rule based classification, Model Evaluation and Selection.			15	CO3
4	Cluster Analysis: Cluster Analysis, Partitioning Methods: K-means clustering; Hierarchical Methods: BIRCH clustering; Density Based Methods: DBSCAN; Grid Based Methods: STING; Data Mining Ethics and Privacy: Ethical considerations in data mining, Privacy-preserving data mining techniques.			15	CO4

Suggested Readings

1. Jiawei Han and Micheline Kamber, "Data Mining Concepts and Techniques" 3rd Edition Elsevier.
2. Pang-Ning Tan, Michael Steinbach and Vipin Kumar, "Introduction to Data Mining", PHI
3. Max Bramer, "Principles of Data Mining", Springer.
4. Arun K Pujari, "Data Mining Techniques", University Press.

Online Resources

1. <https://archive.nptel.ac.in/courses/106/105/106105174/>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	1	1	2	2	2		1	1	1	2	3	1
CO2	2	2	1	1	2	2	2		1	3	2	2	3	1
CO3	3	3	2	3	3	3	2		1	3	1	3	3	3
CO4	3	3	2	3	3	3	2		2	3	2	3	3	3

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	IOT & Technology				
Code	BCAN14212				
Course Type	GE	L	T	P	Credit
Pre-Requisite		3	1	0	4
Course Objectives	To study fundamental concepts of IoT, roles of sensors and hardware in IoT. To learn different Wireless Technologies, protocols for IoT and understand the role of IoT in various domains of Industry.				
Course Outcomes					
CO1	To understand the various concepts, terminologies and architecture of IoT systems.				
CO2	To understand the use of sensors, actuators and IoT supported hardware for design of IoT system.				
CO3	To understand and apply various wireless technology and protocols for design of IoT systems.				
CO4	To understand the various security aspects for IoT system.				
Module	Course Contents			Contact Hrs.	Mapped CO
1	Fundamentals of IoT: Concepts and Definition of IoT, Characteristics, Conceptual Framework, Architectural view, technology behind IoT, M2M Communication; Design Principles for Connected Devices: IoT/M2M systems layers and design standardization, Application of IoT.			15	CO1
2	Hardware for IoT: Sensors, Digital sensors, actuators, radio frequency identification (RFID) technology, wireless sensor networks, participatory sensing technology; Embedded Platforms for IoT: Embedded computing basics, Overview of IoT supported Hardware platforms such as Arduino, Net Arduino, and Raspberry pi.			15	CO2
3	Wireless Technologies for IoT: IEEE 802.15.4, Bluetooth, Wi-Fi, Zigbee, RFID, HART, LoRaWAN, NFCZ-Wave, Z-Wave; IP Based Protocols for IoT: IPv6, 6LowPAN, RPL, REST, AMPQ, CoAP, MQTT.			15	CO3
4	Overview of IoT Security: Introduction Securing the Internet of Things, Architecture, Requirements, Security Protocols for IoT Access Networks, Attack, Defense, and Network Robustness of Internet of Things; Case Studies/Industrial Applications: Home Automation, Smart Cities, Smart Parking, Agriculture and Health Sector, Industrial IoT, Legal challenges, IoT design Ethics, IoT in Environmental Protection.			15	CO4

Suggested Readings

1. Sudip Misra, Anandarup Mukherjee, Arijit Roy "Introduction to IoT" Cambridge University Press; First Edition.
2. Arsheep Bahga, Vijay Madiseti, "Internet of Things: A Hands-On Approach", Orient Blackswan Private Limited - New Delhi; First Edition.
3. Raj Kamal, Internet of Things Architecture and Design Principles, McGraw Hill; Standard Edition.
4. Vibha Soni, "IoT for Beginners: Explore IoT Architecture, Working Principles, IoT Devices, and Various Real IoT Projects", BPB Publications.

Online Resources

1. <https://archive.nptel.ac.in/courses/106/105/106105166/>
2. https://kp.kiit.ac.in/pdf_files/06/SM_6th-Sem_Cse_Internet-of-Things.pdf

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	1	1	1			2	1	1	1	1	1
CO2	2	1	3	1	1	2	1		1	3	1	2	2	1
CO3	1	3	3	2	3	2			1	2	1	2	3	1
CO4	3	3	1	1	1	1	1	1	3	1	3	2	1	2

Program	Bachelor of Computer Applications					
Year	II		Semester		IV	
CourseName	E Commerce					
Code	BCAN14221					
CourseType	DSE		L	T	P	Credit
Pre-Requisite			3	1	0	4
Course Objectives	To provide students with an overview and understanding of E-Commerce with a specific emphasis on Internet Marketing and explore the major issues associated with E-Commerce, security, privacy, intellectual property rights, authentication encryption, acceptable use policies, and legal liabilities. The evolving role of new highly portable, place-aware, always-with-your personal devices in E-Commerce.					
Course Outcomes						
CO1	To Understand the nature and trends in E-Commerce and Mobile commerce					
CO2	To Recognize the business impact and potential of E-Commerce					
CO3	To Explain the technologies required to make E-Commerce viable					
CO4	To Obtain thorough understanding about the security issues, threats and challenges of E-Commerce					
Module	CourseContents				Contact Hrs.	Mapped CO
1	Introduction of E-Commerce: Introduction to Electronic Commerce: E-commerce, Traditional Commerce vs. Ecommerce, Advantages and Disadvantages of E-commerce, Impact of E-commerce, Classification of E-commerce, Applications of E-commerce, Limitations of E-commerce, Emerging Trends in E-commerce, Electronic Commerce Business Models: Native Content based Model, Transplanted Content Model, Native Transaction Model, Transplanted Transaction Models.				15	CO1
2	E-Commerce Framework and Business Security: Architectural Framework of Electronic Commerce, Network Infrastructure, Information Distribution Technology, Networked Multimedia Content Publishing Technology, Security and Encryption, Payment Services, Business Service Infrastructure, Public Policy and Legal Infrastructure, Vulnerability of Information on Internet: Security Policy, Procedures and Practices, Site Security, Protecting the Network, Firewalls.				15	CO2
3	Electronic Commerce Network Security & Payment System: Transaction Security, Cryptology, Digital Signatures, Electronic Mail Security, and Security Protocols for Web Commerce, Electronic Payment System. Introduction to Payment Systems: Online Payment Systems, Pre-Paid Electronic Payment Systems, Postpaid, E-Security: Information system Security, Security on the Internet, E-Business Risk Management Issues Information Security Environment in India. Legal and Ethical Issues :Cyber talking-Privacy is at Risk in the Internet Age, Phishing, Application Fraud , Skimming , Copyright-Internet Gambling , Threats to Children.				15	CO3
4	Mobile Commerce: Introduction and Applications, Mobile Commerce, Benefits of Mobile Commerce, Impediments in Mobile Commerce, Mobile Commerce Payment Systems, Mobile Commerce Applications, Case Study: Future of E-commerce in India, Flipkart: Journey of E- Commerce Start-Up.				15	CO4

Suggested Reading

1. Bharat Bhaskar, "Electronic Commerce: Framework, Technologies & Applications", TMH
2. David Whiteley, "E-Commerce Strategy, Technologies and Applications", Tata McGraw Hill
3. Ravi Kalakota, Andrew Winston, "Frontiers of Electronic Commerce", AddisonWesley
4. Bajaj and Nag, "E-Commerce: The Cutting Edge of Business", Tata McGraw Hill
5. P. T. Joseph, "E-Commerce: An Indian Perspective", PHI Learning Pvt. Ltd.

Online Resources:

1. https://onlinecourses.swayam2.ac.in/cec19_cm01/preview
2. https://onlinecourses.swayam2.ac.in/nou21_cm14/preview

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2		2		2	1	1		1			2	1	1
CO2	2		2		2	2	1		1			2	1	1
CO3	2	2	2	1	3	2	1		2	2	2	2	2	2
CO4	2	2	3	1	2	2	1		1	2	2	2	3	3

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	E-Governance				
Code	BCAN14222				
Course Type	DSE	L	T	P	Credit
Pre-Requisite		3	1	0	4
Course Objectives	The course aims to provide students with the comprehensive understanding of the concept of E-Governance and to know different e-governance models and infrastructure development. The use data warehousing and mining in e-governance and to understand the role of E Governance in various sectors.				
Course Outcomes					
CO1	To understand the principles of E-Governance and its model.				
CO2	To develop skills to critically evaluate E-readiness and strategies				
CO3	To understand the applications in Data Mining.				
CO4	To educate E-Governance with its use cases in various sectors.				
Module	Course Contents			Contact Hrs.	Mapped CO
1	Overview of E-Governance and its Models: Introduction to E-Governance, Needs of E-Governance, Issues in E-Governance Applications and Digital Divide; Evolution of E-Governance, its scope and content; Present global trends of growth in E-Governance; Evolution in E-Governance and Maturity Levels: Five Maturity Levels; Characteristics of Maturity Levels; Key areas.			15	CO1
2	E-Governance Infrastructure, Strategies: E-Government Models; E-readiness; Digital System Infrastructure, Legal Infrastructural Preparedness, Institutional Infrastructural Preparedness, Human Infrastructural Preparedness, Technological Infrastructural Preparedness; Evolutionary Stages in E-Governance.			15	CO2
3	Data Warehousing and Data Mining in Government: Features of Data Warehouse and data Mining; Significance of Data Warehouse: Census Data, Prices of Essential Commodities; Other areas for Data Warehousing and Data Mining: Agriculture, Rural Development, Health, Planning, Education, Commerce and Trade, Other Sectors.			15	CO3

4	Case Studies of E-Governance in Indian perspective NICNET: Role of Nationwide Networking in E-Governance, Smart Nagarpalika-Computerization of Urban Local Bodies (Municipalities, EkalSeva Kendra, Aadhar, E-Suvidha, Bhulekh).	15	CO4
---	--	----	-----

Suggested Readings

1. C.S.R. Prabhu, "E-Governance: Concepts and Case Studies", Prentice-Hall of India Private Limited.
2. N. Gopalsamy, "Information Technology & e-Governance", New Age Publication, First Edition.
3. Backus, Michael, "e-Governance in Developing Countries", IICD Research Brief.
4. SubhashBhatnagar, "Unlocking E-Government Potential: Concepts, Cases and Practice, SAGE Publications India Pvt Ltd.

OnlineResources

1. <https://nptel.ac.in/courses/129106001>
2. <http://acl.digimat.in/nptel/courses/video/106106093/L32.html>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	2		2	3	1		1			2	1	1
CO2	1	1	2		2	2	2		1			2	1	1
CO3	2	2	2	2	3	2	3		2	2	2	2	2	2
CO4	2	2	3	2	2	2	3		3	2	2	2	3	3

Program	Bachelor of Computer Applications					
Year	II	Semester			IV	
CourseName	Enterprise Resource Planning					
Code	BCAN14223					
Course Type	DSE	L	T	P	Credit	
Pre-Requisite		3	1	0	4	
Course Objectives	To present the fundamental concepts of ERP. To provide a contemporary approach towards ERP systems and integrating business process across functional areas. To make the students aware regarding the need and importance of ERP in today's scenario. To train the students to develop the basic understanding of how ERP enriches the business organizations in achieving a multidimensional growth. To develop understandings about the concept like e-ERP, e-CRM, e-SCM.					
Course Outcomes						
CO1	Able to Introduce ERP system, its structural representation, evolution along with MRP, MIS, EIS, DSS, DWDM.					
CO2	Able to classify ERP business models, their working and Introduction to BPR, SCM and CRM.					
CO3	Able to identify various ERP modules and Implementation of different phases, life cycle, methodologies of ERP.					
CO4	To understand Interdependence of ERP and E-Commerce. Concept of ERP markets, role of vendors and Consultants in ERP Implementation.					
Module	CourseContents				Contact Hrs.	Mapped CO
1	Introduction: Basic Concept of ERP System; Enterprise – An overview, Need of ERP; Structure of ERP; Common Misconception about ERP; Benefits & Risk of ERP. Evolution of ERP: Concept of Material Requirement Planning (MRP), Manufacturing Resource Planning (MRP II); Related Technologies: MIS (Management Information System), EIS (Executive Information System), DSS (Decision Support System), Role and Importance of Data Warehouse and Data Mining in ERP.				15	CO1
2	ERP Business Modelling: Building the Business Model, Role of Consultants, Vendors and Users – Customization – Precautions – ERP: Post-implementation Options – ERP Implementation Methodology – Guidelines for ERP Implementation. Introduction to BPR, SCM and CRM Energy: Business Process Reengineering: ERP and BPR, Supply Chain Management (SCM): Evolution, Components, and Characteristics of SCM, ERP and SCM; Role and Need of CRM, Integration of ERP, SCM and CRM.				15	CO2
3	ERP Modules: Introduction, Finance, Production, Marketing, Quality Management, Logistics, Material Management, Integration of ERP, Supply chain and Customer Relationship Application.ERP Implementation: Implementation challenges, ERP implementation, Strategies, ERP Implementation Lifecycle, Implementation Methodologies, ERP project teams, vendors, consultants, Dealing with employee resistance, training and education, data migration, Project monitoring and management,				15	CO3

	Post implementation activities.		
4	ERP Market: ERP market place, Market Place Dynamics, SAP, BANN, JD Edwards, PeopleSoft, Oracle, Just in Time, Make-To-Order, Make- To-Stock. Future Directions Different Roles in ERP Implementation: Role, Need, Evaluation Criteria for Vendors, Role and Need of ERP Consultants, Role of End Users, ERP and E-commerce, Future Directions in ERP, Concept of e-ERP, e-CRM, e-SCM	15	CO4

Suggested Readings-

1. Alexis Leon, "Enterprise Resource Planning Demystified", Tata McGraw-Hill Publishing Company Ltd.
2. Vinod Kumar Garg and N.K. Venkitakrishnan, "Enterprise Resource Planning– Concepts and Practice", Prentice Hall of India.
3. Rahul V. Altekhar, "Enterprisewide Resource Planning", Tata McGraw Hill.
4. Mary Summer, "Enterprise Resource Planning", Pearson Education

Online Resource-

1. <https://www.coursera.org/articles/what-is-erp>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	1	2				2	2	1	2	2	2
CO2	2	2	2	1	2	1			2	2	1	2	2	2
CO3	2	2	2	1	2	1			1	2	1	1	2	2
CO4	2	1	2		2						1		2	2

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	Python Programming Lab				
Code	BCAN14251				
Course Type	DSC	L	T	P	Credit
Pre-Requisite		0	0	4	2
Course Objectives	It provides the practical implementation of developing and debugging Python Programs using core data structures like Lists, Dictionaries, Tuples, and Strings as well as understand the concept of functions, modules and NumPy Library.				
Course Outcomes					
CO1	Acquire programming skills in core Python using various programming constructs, functions, and strings.				
CO2	Implement methods to create and manipulate lists, tuples and dictionaries and basic introduction of NumPy.				
Module	Course Contents			Contact Hrs.	Mapped CO
1	1. Installing and configuring Python on windows, Linux. 2. Introduction to Variables, keywords, basics operation in python, Taking input in console. 3. Taking multiple inputs from user in Python operators' implementation 4. Displaying Output using print () function, using end parameter in print (), 5. Practical implementation of the constructs like if, else, elseif ladder. 6. Implementation of range function in for loop. 7. Implementation of Special keyword - in and is. 8. Implementation of looping constructs using for loop, range() function and examples use of enumerate, zip() function. 9. Implementation of strings in python, single quoted/double quoted/triple quoted Strings, string functions - split, trim, join, format, replace, count, find, index, rjust, ljust, center, upper, lower. Note: - Students will also perform all other exercises provided by course instructor.			30	CO1
2	1. Practical implementation of list, creation and traversal, list functions - append, insert, extend, remove, pop, clear, sort, count, index, copy. 2. Practical implementation of tuples, creation, and traversal. 3. Practical implementation of Set, creation, and traversal, set functions - add, update, remove, clear, pop, union, intersection, difference, disjoint, subset, superset. 4. Practical implementation of Dictionary, creation and traversal. 5. Practical implementation of dictionary function - get, update, keys, items, values. 6. Making module for functions and importing them different types of imports in python.			30	CO2

	7. Practical implementation of 1d array and its attributes. 8. Practical implementation of 2d array and its attributes. Note: - Students will also perform all other exercises provided by course instructor.		
--	---	--	--

Suggested Readings

1. Kenneth A. Lambert, the Fundamentals of Python: First Programs, Cengage Learning.
2. Practical Programming: An introduction to Computer Science Using Python, second edition, Paul Gries, Jennifer Campbell, Jason Montojo, The Pragmatic Bookshelf.
3. NumPy for Beginners: first Step to Learn to Data Science – Preeti Saraswat .
4. David Beazley, Brian K. Jones “Python Cookbook”, 3rd Edition. O’Reilly Publications.

Online Resources

1. <https://nptel.ac.in/courses/106106145>
2. <https://www.python.org/>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2		2	1	2		1			2	1	1
CO2	2	1	2		2	2	2		1			2	1	1

Program	Bachelor of Computer Applications				
Year	II	Semester		IV	
Course Name	.NET FRAMEWORK & C# LAB				
Code	BCAN14252				
Course Type	DSC	L	T	P	Credit
Pre-Requisite		0	0	4	2
Course Objectives	To present the fundamental concepts of Windows Desktop and Website development through Microsoft Technologies. To impart a solid foundation and develop the skill of Web Development through C# Programming. To develop the concepts of static and dynamic Web Pages and make students familiar with Client servers to learn about technology, Distributed Applications and Web Services.				
Course Outcomes					
CO1	To work on the fundamental concepts of Windows Desktop and Website development through Microsoft Technologies.				
CO2	To become able to develop the skill of Web Development through C# Programming. And develop the concepts of static and dynamic Web Pages and make the students familiar with web services.				
Module	Course Contents			Contact Hrs.	Mapped CO
1	1. Implementation of Decision Making and Branching Statements on Console Applications. 2. Implementation of Iterative Statements on Console Applications. 3. Implementation of Arrays and Array List on Console Applications. 4. Implementation of Boxing and Unboxing on Console Applications. 5. Implementation of Strings on Console Applications. 6. Implementation of Inheritance and Polymorphism on Console Applications. 7. Implementation of Interfaces on Console Applications. 8. Implementation of Properties and Indexers on Console Applications. Note: Students will also perform all other exercises provided by course instructor.			30	CO1
2	1. Implementation of Multithreading in C# 2. Implementation of private assemblies in .NET Applications. 3. Implementation of shared assemblies in .NET Applications. 4. Implementation of Server-Side Controls in asp.net. 5. Implementation of Database Connectivity in asp.Net 6. Implementation of various Data Rendering Controls in asp.Net. 7. Implementation of Web Services in ASP.Net Applications. 8. Construct the C# console application to implement the Operator Overloading. Note: Students will also perform all other exercises provided by course instructor.			30	CO2

Suggested Readings

1. Balagurusamy, "Programming. with C#", Tata McGraw Hill Publication.
2. Stephen C. Perry, Atul Kahae, Stephen Walther, Joseph Mayo," Essential of .NET and Related Technologies with a focus on C#, XML, ASP.net and ADO.net", Pearson.
3. Jospeh Albahari, "C# 8.0 Pocket Reference", O'Reilly.

Online Resources

1. <https://archive.nptel.ac.in/courses/201/202/203/>
2. <https://archive.nptel.ac.in/courses/306/309/310/>

Course Articulation Matrix														
PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2		2		2	1	2		1			2	1	1
CO2	2		2		2	2	2		1			2	1	1